

Advanced Macroeconomics

Calibration and the use of Data in Macroeconomics

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Introduction

- A discussion on the use of data in macro models and empirical strategies in macro.
- Most of the discussion here is my own thoughts on the topic (that changes constantly over time) and mostly applied to het. agents models.
 - ▶ Some material came from the notes of Matthias Kredler.

References

- Nakamura and Steinsson (2018, JEP): Modern review of the “Identification in Macro”.
- Canova (2007, Book), Fernández-Villaverde et al (2009): Great references for estimation in Macro. Chapter 7 of Canova has a very nice discussion on “calibration”.
- Chodorow-Reich (2020, JEDC), Guren et al. (2021, NBER Macro). Good references for the cross-regional empirical methods. Check Chodorow-Reich class [notes](#) as well.

- What is a model?
 - ▶ **Sargent**: define a model as a probability distribution over a sequence of outcomes, possibly indexed by a parameter vector;
 - ▶ **Wikipedia**: a theoretical construct representing economic processes by a set of variables and a set of logical and/or quantitative relationships between them.
 - ▶ **Statistics**: a data generating process.
- Why use models?
- We are usually interested in *counterfactuals*: i.e. what happens to Y when we change X ?
 - ▶ **Applied Micro**: counterfactual comes from natural experiments + statistical assumptions.
 - ▶ **Quantitative Macro**: counterfactual comes from a set of structural assumptions using economic theory and the parameters chosen for the models.
- Must choose a set of parameters θ to make counterfactuals.

Choosing parameters

- How we identify parameter values often matters a great deal for results in quantitative work.
- A method that is (or tries to be) fully transparent and has a cookbook recipe: maximum-likelihood.
- Another approach is to use some type of moment-based estimation, i.e. GMM, minimum distance.
- Often authors refer to choosing parameters as **calibration**. What is the difference?

Likelihood based

- Systematically uses all moments (full information).
- Used for DSGE models $\Rightarrow$ few data points!
- Use Bayesian estimation to make feasible.
- But: Have to choose which data series to match $\Rightarrow$ Similar issue arises as for moment-matching – what to choose?

- What calibration might mean (from Canova's book):
 - ▶ "...one wants to calibrate a model (in the sense of selecting reasonable parameters values) because there is no data to estimate its parameters."
 - ▶ "...one may prefer to calibrate (as opposed to estimating it) if the misspecification is so large that statistical estimation of its parameters will produce inconsistent and/or unreasonable estimates and formal statistical testing will lead to outright rejection."
 - ▶ "...some users interpret calibration as an econometric technique where the parameters are estimated using "economic", as opposed to "statistical", criteria."
- "the term calibration is used to indicate a particular collection of procedures designed to provide an answer to economic questions using "false" models."

Calibration

The essence of calibration by **Kydland and Prescott (1991, 1996)**:

- (i) Choose an economic question to be addressed.
- (ii) Select a model design which bears some relevance to the question asked.
- (iii) Choose functional forms for the primitives of the model and find a solution for the endogenous variables in terms of the exogenous ones and of the parameters.
- (iv) Select parameters and convenient specifications for the exogenous processes and simulate paths for the endogenous variables.
- (v) Evaluate the quality of the model by comparing its outcomes to a set of "stylized facts" of the actual data.
- (vi) Propose an answer to the question, characterize the uncertainty surrounding the answer and do policy analyses if required.

- Still many questions:
 - ▶ What are the set of “stylized facts”?
 - ▶ What is the measure of distance used to compare the model with actual data?
- A lot of discretionary choices by the research.
- Here it lies important philosophical aspect of the methodology:
 - ▶ In a strict sense, all models are approximations to the DGP and, as such, false and unrealistic;
 - ▶ Once this point of view is accepted, it makes no sense to examine the validity of a model using standard statistical tools which assume it to be true, at least under the null.
 - ▶ This is what is implicitly assumed with GMM and ML.

- The modern definition of calibration can be summarized by this set of [tweets](#) from Jon Steinsson.
- **Jon Steinsson**: “Calibration is just moment matching without standard errors.”
- Std. errors are important, but parameter uncertainty is minor relative to model misspecification.
 - ▶ Sometimes it is too computationally costly to calculate standard errors (especially in heterogeneous-agents economies).
 - ▶ Not everybody agrees with this point.
- Model evaluation by calibration gives rise to “portable statistics”, i.e., statistics that are used over and over again to evaluate different models.

- The moments are generated after a full solution of the model (as opposed to only one FOC or equation).
 - ▶ In some sense, this type of moment matching is similar to ML since requires solving the full model.
 - ▶ Calibration gives the freedom to choose which moments to match, in ML the freedom is selecting the set of observables to use for estimation.
- Nowadays, we are moving from moments that are just averages/correlations/variances, to matching causal estimates:
 - ▶ e.g., marginal propensity to consume out of a transitory fiscal rebate;

Models as over-identifying Restrictions

- A good theory imposes restrictions on the data.
- These restrictions/predictions can be tested and falsified.
- **Example:** Have normal distributions of body height for n countries: $\{\hat{\mu}, \hat{\sigma}\}_{i=1}^n$
- The model $x_i \sim N(\mu_i, \sigma_i)$ with $2n$ parameters is not particularly interesting: no degrees of freedom!
- Better: Posit that μ_i is a function of some covariate at country level: $\mu_i = \alpha x_i$, removes $n - 1$ parameters.
- **Perfect theory:** Few parameters, but can match all moments.

The name of the game (usually)

- Calibrate m free model parameters to m targets.
- Then show k additional (non-targeted) moments in model and data: “model validation”.
- Fit should be decent to have a good model.
- **Example:** Target levels of an economy in 2000, then see how model does for period 2000-2020.
- Alternative (not done much): Could choose m moments to minimize distance to $m + k$ moments.
 - ▶ Perfect model: Cannot reject over-identifying restrictions statistically $\Rightarrow$ almost impossible in practice in economics!
 - ▶ OK model: We feel the quantitative economic fit of the moments is satisfactory to use the model to do policy evaluation etc.

Calibration in Practice

- (i) Calibrate parameters before solving the model (“outside the model”):
 - ▶ Estimate some parameters using data without having to solve the model: Earnings/productivity process, progressive taxation/transfers.
 - ▶ Take (uncontroversial) parameters from other studies.

- (ii) Calibrate the remaining parameters to match moments (exactly or over-identified method of moments).
 - ▶ In HA models, we usually target micro moments in the steady state of model: wealth distribution, marginal propensities to consume, firm size distribution.
 - ▶ If your model have transitions/cycles you can target some macro moments (time-series) after you calibrated the steady state.

- (iii) Validate your model using non-target moments.

Step 1: Parameters directly identified outside model

Examples:

- Processes that the model takes as exogenous and that we can estimate from data:
 - ▶ Income process in heterogeneous-agents model.
 - ▶ TFP in business cycles models.
 - ▶ Exogenous process for exit in firm dynamics models.
- Variables that can be taken from institutional environment/prices:
 - ▶ tax rates, tax schedules, tariffs.
 - ▶ relative prices of goods pin down technology in simple settings: $y = An$.

Step 2: Take parameters from other studies

- **Kydland and Prescott (1991)**: choosing parameters using information obtained from other studies imposes coherence among various branches of the profession.
- Prime example: CES-utility curvature parameter ($\gamma = 2$).
- But careful: Does the same Greek letter really mean the same in my model and the other? Often not!
- **Example**: Risk aversion
 - ▶ measured in lab experiment with small sums;
 - ▶ risk aversion coefficient of representative agent in RBC model.
- A good compromise is to test the robustness of your results in an interval of these parameters consistent with empirical estimates.
 - ▶ e.g.: $\gamma \in [0.5, 1, 2, 3]$.

Step 3: Moment-matching

- What is often done: Minimize percentage differences.

$$\min_{\theta} \sum_{i=1}^n W_i [\ln \mu_i(\theta; \psi) - \ln \hat{\mu}]^2$$

where ψ is a vector of “deep parameters” chosen in the previous step and W_i some arbitrary weight.

- ▶ resembles usual moment-based estimators like SMM;
 - ▶ in fact, if we use some statistical criteria to minimize the loss function (i.e., minimize MSE, choose W_i efficiently, etc), the two methods are exactly the same;
 - ▶ the difference is in the discretionary choice of the calibrator;
 - ▶ **Canova**: “...a calibrator may look like an econometrician who uses different loss functions in different parts of the model”
 - ▶ “...a calibrator may also look like an inefficient GMM econometrician.”
- **Important**: the parameters θ are *conditional* on ψ .

Detour: Indirect Inference

- Another approach similar to SMM is **indirect inference**.
 - ▶ Simulate model data;
 - ▶ Estimate an auxiliary model (e.g., a linear regression) in both the simulated and actual data;
 - ▶ Minimize the distance between the coefficients of the **auxiliary model**.
- Useful when the structural relationships are difficult to express as simple unconditional moments.
- **Examples:**
 - ▶ Guvenen and Smith (2014, ECTA): Consumption-savings with uncertainty about income-process.
 - ▶ Search-friction models with wage dispersion (e.g., Lise (2013, ReStud)).
- How to choose the auxiliary model?

Step 3: Moment-matching

Practical Issues:

- Hard task computationally
 - ▶ If moments smooth in θ : Use gradient-based method.
 - ▶ Otherwise: Use more robust methods (simplex, genetic algorithms, etc).
 - ▶ If too slow/impossible: try to understand how parameters change the moments generate by the model.
- **Key challenge:** Which moments to pick?
 - ▶ informative moments: Which statistics are especially affected by a certain parameter?
 - ▶ Often hardest, but most important: elasticity-type (curvature) parameters. (labor-supply elasticity, risk aversion,...)

Informative moments

- Before estimating: Draw comparative-statics graphs with your model.
- Which parameter affects which moment most? Use this moment to pin down the parameter.
- Can also choose by economic reasoning on model properties.
- Good papers have discussion on identification (usually no proof since no direct one-to-one match).

Elasticity/curvature moments

- Determines how much agents change behavior when prices/incentives change.
- Can use cross-sectional distribution to pin this down.
- Ríos-Rull: “Don’t identify elasticity/slope moment by a level moment”.
- **Causal Effects as Identified Moments**
 - ▶ Target a moment “causally” estimated by the applied micro literature;
 - ▶ Target an impulse response function.
 - ▶ The advantage of identified moments is that they can provide evidence on specific causal mechanisms of a model and may be relatively invariant to other model features.

Other useful tricks

- Re-parameterize your model to have an unconstrained optimization problem. Examples:
- $\sigma = \exp(\hat{\sigma})$ for $\sigma > 0$ or $\beta = \text{logist}(\hat{\beta})$ for $\beta \in (0, 1) \Rightarrow \hat{\sigma}$ and $\hat{\beta}$ live on entire real line
- Include equilibrium conditions in the loss function.
 - ▶ Example: Net demand of assets must be zero in equilibrium.
 - ▶ Penalize (net demand)²; put harsher penalties than on other moments.
 - ▶ Useful when equilibrium-finding loop is computationally costly.
- Smooth discrete choices by adding preference shocks
 - ▶ Choice probabilities instead of 0-1.
 - ▶ Makes moments smoother in parameters.

Step 4: Model validation

- There are no free parameters and no uncertainty is allowed, so how to validate the model?
- Typical approach: use non-target moments
- **Examples:**
 - ▶ Kaplan-Moll-Violante (HANK): calibrate share of hand-to-mouth, validate using moments of wealth distribution;
 - ▶ Midrigan-Xu: calibrate productivity process, validate using autocorrelations of investment and employment.
- Bonus (hard) approach: replicate empirical studies!
 - ▶ Berger-Herkenhoff-Mongey (2022, AER): GE model of firms' labor market power.
 - ▶ Replicate “natural experiments” from empirical papers.

Step 4: Model validation

- It is useful to assess whether the moments are able to identify the parameters.
- A simple test is to “perturb” a parameter and compute how “sensible” is the loss function to changes in parameters.
 - ▶ Perturb each parameter (one at a time) by 1% (or any small value);
 - ▶ Compute the % change of the loss function relative to the value evaluated at the “estimated parameters”
- if the model is well identified, the loss function should not be flat in the region around the vector of estimated parameters.
- You can even go further and compute the % change for the contribution of **each moment** to the total loss.

Estimating useful parameters of HA models:
stochastic processes, distributions, and more.

Stochastic Process

- The first, and perhaps one of the most important parameters in HA models are the parameters of the earnings/productivity process.
- In the simplest version, the stochastic process is given by:

$$y_{it} = \rho y_{it-1} + \varepsilon_{it}, \quad \varepsilon_t \sim N(0, \sigma^2) \quad (1)$$

- To estimate this process you need panel data of income/productivity of at least 2-periods so you have enough information on the persistence (ρ) and inequality (σ^2).
- As it will become clear later, this is a simple process, more involved processes will require more information - either higher moments or longer time series.

Intuition Stochastic Process

- Note that there are two settings of moments informative about the process, the variance of earnings/productivity in levels, $V(y_{it})$, $V(y_{it-1})$ and in growth $V(\Delta y_{it})$, where $\Delta y_{it} = y_{it} - y_{it-1}$

- Taking the variance in equation (1)

$$V(y_{it}) = \rho^2 V(y_{it-1}) + \sigma^2$$

- Subtracting y_{t-1} from both sides and taking the variance in equation (1):

$$y_{it} - y_{t-1} = (\rho - 1)y_{it-1} + \varepsilon_{it},$$

$$V(\Delta y_{it}) = (\rho - 1)^2 V(y_{it-1}) + \sigma^2$$

Basic Stochastic Process

- You have 2 equations, 2 unknowns (ρ, σ^2) , and three moments $V(y_{it})$, $V(y_{it-1})$, $V(\Delta y_{it})$
 $\Rightarrow$ overidentified system:

$$\begin{aligned}V(y_{it}) &= \rho^2 V(y_{it-1}) + \sigma^2 \\V(\Delta y_{it}) &= (\rho - 1)^2 V(y_{it-1}) + \sigma^2\end{aligned}$$

- If you assume the system is stationary you can use either $V(y_{it-1})$ or $V(y_{it})$ (in infinite horizon).
- In life-cycle models, you could use the extra moment to identify initial heterogeneity.
- Note that you can substitute $V(\Delta y_{it})$ by the autocovariance $C(y_{it}, y_{it-1})$.

Transitory-persistent Process

- A popular alternative is to model the earnings/productivity process as the sum of a transitory and a persistent component:

$$y_{it} = z_{it} + \varepsilon_{it}$$

$$z_{it} = \rho z_{it-1} + \eta_{it}$$

where $\varepsilon_{it} \sim N(0, \sigma_\varepsilon^2)$ is the shock of the transitory component, and $\eta_{it} \sim N(0, \sigma_\eta^2)$ the shock of the persistent component.

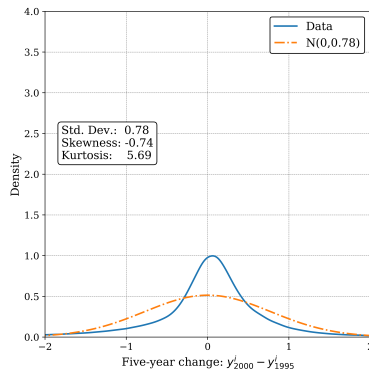
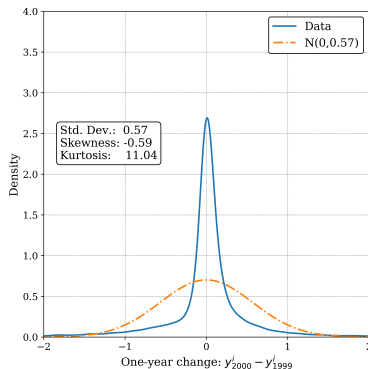
- ▶ **Transitory:** Bonus, health shocks, short unemployment spells
 - ▶ **Persistent:** Promotions, unemployment spells with scarring effects.
- Persistent shocks matter more for welfare and savings behavior. In firm dynamic models, persistent productivity interacts with adjustment costs or financial frictions.

Transitory-persistent Process

- The transitory-persistent provides better fit and captures the income dynamics of longer horizon.
- Requires **at least four periods** of panel data.
- It can still be discretized using the usual methods, but the state space increases fast.
- Identification requires the autocovariance matrix of earnings (in growth rate or in levels). Estimation usually done using minimum distance/GMM.
- More information on the econometric identification: Guvenen (RED, 2009) identification in levels; Blundell, Pistaferri and Preston (AER, 2008) identification in growth.

Higher moments of earnings growth

- Guvenen et al (ECTA, 2021) emphasizes the role of higher-moments, non-linearities and age-dependence of earnings growth.



- This is also true in Brazil.

Stochastic process with higher moments

- Higher moments give additional incentives for precautionary savings. We can specify the earnings processes with higher-moments:

$$y_{it} = z_{it} + \varepsilon_{it},$$

$$z_{it} = z_{it-1} + \eta_{it},$$

$$\eta_{it} \sim \begin{cases} N(\mu_{\eta,1}, \sigma_{\eta,1}^2) & \text{with prob. } p_{\eta} \\ N(\mu_{\eta,2}, \sigma_{\eta,2}^2) & \text{with prob. } 1 - p_{\eta} \end{cases}$$

$$\varepsilon_{it} \sim \begin{cases} N(\mu_{\varepsilon,1}, \sigma_{\varepsilon,1}^2) & \text{with prob. } p_{\varepsilon}, \\ N(\mu_{\varepsilon,2}, \sigma_{\varepsilon,2}^2) & \text{with prob. } 1 - p_{\varepsilon}. \end{cases}$$

where the shocks are drawn from a mixture of normals. Other distributions are also possible.

Stochastic process with higher moments

- Still requires long panel data and specially you must feed **higher moments** of the distribution in the estimation of the extra parameters $(p_{\eta}, p_{\varepsilon}, \dots)$.
- Luckily, the moments of the earnings growth distribution (for Brazil) are available in the GRID project: <https://www.grid-database.org/>.
- You must be careful and think whether your moments identify the higher moments.
- Estimation is usually done through simulated methods of moments (SMM). It is slow, but it is done outside of the model.
- Discretization is not trivial, but can be done relatively fast using simulation methods. The reference is: DiNardi et al (JEEA, 2020).

Other approaches and Extra Issues

- You can combine other shocks in the stochastic process to capture different dimensions not captured by income:
 - ▶ **Unemployment:** with some exogenous probability the agent becomes unemployed.
 - ▶ **Superstar shock/entrepreneurs:** with some exogenous probability the agent becomes an entrepreneur (Castañeda et al, 2003; Bayer and Luetticke many papers).
- What other features could be incorporated?
- **Business cycles:** There is a large literature on the cyclicalities of risk, including higher moments.
 - ▶ HA literature knows that this matters for precautionary savings and consumption (McKay, JME, 2017) but still relatively unexplored in HANK (exception is Bayer et al. ECTA, 2019).

Productivity Process

- Earnings are observable, but usually productivity is not. To estimate the productivity process one must first estimate productivity or infer from either employment or sales data.
- The usual method to retrieve firm-level TFP requires production function estimation:

$$Y_{jst} = \exp(z_{jst}) K_{jst}^{\alpha_s} L_{jst}^{\beta_s}$$

where j is firm, s is sector and t time. Y_{jst} can be either sales or physical output.

- ▶ Take the logs, add an error term ε and you have an equation to estimate the parameters.
- ▶ The IO people worked really hard on the identification of this equation: Olley and Pakes (1996), Levinsohn and Petrin (2003), Akerberg et al. (2015), De Loecker and Syverson (2021).
- ▶ Under some assumptions about the inputs, you can also retrieve the markups (De Loecker and Warzynski, 2012; De Loecker and Eeckhout, 2020) and even markdowns (Yeh et al, 2022).
- ▶ Usually requires firm-level panel data, but with very strong assumptions you are able to identify using a cross-section (i.e., cost share approach).

Productivity Process

- If you cannot estimate the production function, you can try to recover the productivity process using employment or sales data.
- Recall that in the basic Hopenhayn model, there is a one-to-one map between productivity and labor demand:

$$n = \left(\frac{\alpha z}{w} \right)^{\frac{1}{1-\alpha}},$$

so movements in $\ln z$ correlate with movements in $\ln n$.

- You have to be careful about things that break this correlation: labor (and capital) adjustment cost, changes in firm-specific wedges, etc.
- See Sterk, Sedláček and Pugsley (2021).

Firm Size Distribution

- The firm size distribution in het. firms model is closely related to the productivity distribution.
- Calibration typically involves a combination of: dispersion of shock, distribution of entrants, fixed costs, entry costs.
- Additional moments from firm dynamics should also be targeted, including: Entry and exit rates, labor reallocation, etc.
 - ▶ If you have adjustment costs, you can use moments such as the fraction of inaction as targets.
- Typically, relative moments (e.g., size of entrants relative to average size) are targeted, but sometimes you want the actual firm size in number of workers.
 - ▶ Use aggregate parameters such as a labor supply shifter or aggregate productivity to align average firm size (the level) with data.

Progressive Taxation

- Idiosyncratic shocks imply earnings inequality. Progressive taxation matters, since it redistributes from the top to the bottom: changes wealth distribution, MPCs, etc.
- Suppose the tax function has the following form:

$$y_i^n = F(y_i),$$

where y_i^n is net income and y_i is gross income.

- What function should we use? Two approaches:
 - ▶ Log-linear form;
 - ▶ Brackets;

Log-linear Form

- A functional form that captures progressivity (See Benabou (2002), Heathcote et al. (2017)):

$$T(y) = y - \tau_1 y^{1-\tau_2} \quad \text{where } y \text{ is the individual gross labor income.}$$

- ▶ τ_2 gives the degree of progressivity, i.e. it measures the elasticity of posttax to pretax income.
 - ▶ Given τ_2 , τ_1 shifts the tax function and determines the average level of taxation in the economy.
- This implies that map from gross income to net income is:

$$y_i^n = F(y_i) = y_i - T(y_i) = \tau_1 y_i^{1-\tau_2}$$

- Parameters can be easily estimated in regressing $\log y_i^n$ on $\log y_i$.

Log-linear Form

- The tax is progressive if the ratio of marginal to average tax rates is larger than 1 for every level of income.
 - ▶ $\tau_2 = 1$: full redistribution $\Rightarrow T(y) = y - \tau_1$.
 - ▶ $0 < \tau_2 < 1$: progressivity $\Rightarrow T'(y) > \frac{T(y)}{y}$.
 - ▶ $\tau_2 = 0$: no redistribution $\Rightarrow T'(y) = \frac{T(y)}{y} = 1 - \tau_1$.
 - ▶ $\tau_2 < 0$: regressivity $\Rightarrow T'(y) < \frac{T(y)}{y}$.
- Break-even income: $y_{be} = \tau_1^{\frac{1}{\tau_2}}$.
 - ▶ If $y_i > y_{be}$, i is a taxpayer.
 - ▶ If $y_i < y_{be}$, i receives a transfer.

Progressive Taxation

- **Log-linear:**
 - ▶ **Good:** Flexible; Easy to estimate if you have the data.
 - ▶ **Bad:** Cannot account for specific marginal rates; Cannot be estimated if you do not have gross and net income for the same i (in the US they input using TAXSIM).
- Alternative: replicate the actual tax system in the function F .
- Include brackets of all marginal rates, but also possible transfers. **Brackets:**
 - ▶ **Good:** Account for top marginal rates. Very flexible.
 - ▶ **Bad:** How to model the **entire** transfer system? What to include and what to leave out?

Wealth Distribution

- Getting a “correct” wealth distribution was at the core of the early literature of heterogeneous agents.
- Early papers → getting the top right
- Various approaches (see DiNardi and Fella, RED, 2017):
 - ▶ Correct income process;
 - ▶ Preference heterogeneity;
 - ▶ Life-cycle motives: bequest, human capital, health shocks;
 - ▶ Entrepreneurship.
 - ▶ Heterogeneity (and shocks) in r_t .
- HANK papers → getting the bottom right → getting the right MPC (core mechanism of transmission of aggregate shocks).

Wealth Distribution

- Which moments to target?
- **Example:** Kaplan, Moll and Violante:

TABLE 5

	Data	Model	Moment	Liquid wealth		Illiquid wealth	
				Data	Model	Data	Model
Mean illiquid assets	2.92	2.92	Top 0.1 percent share	17	2.3	12	7
Mean liquid assets	0.26	0.23	Top 1 percent share	47	18	33	40
Frac. with $b = 0$ and $a = 0$	0.10	0.10	Top 10 percent share	86	75	70	88
Frac. with $b = 0$ and $a > 0$	0.20	0.19	Bottom 50 percent share	-4	-3	3	0.1
Frac. with $b < 0$	0.15	0.15	Bottom 25 percent share	-5	-3	0	0
Gini coefficient				0.98	0.86	0.81	0.82

Notes: Left panel: moments targeted in calibration and reproduced by the model. Means are expressed as ratios to annual output. Right panel: statistics for the top and bottom of the wealth distribution not targeted in the calibration.

Source: SCF 2004

Calibrating the Wealth Distribution

- Early approach: permanent heterogeneity in β (Krussel-Smith, 1998).
- For instance, suppose: $\beta \in [\bar{\beta} - \epsilon, \bar{\beta} + \epsilon]$.
- Discretize the space of β with uniform probability (Krueger, Mitman and Perri, 2016).
- You can also calibrate the beta of each group g individually: β targeting specific moments of the percentiles of the wealth distribution.
- Then $\bar{\beta}$ to match wealth-to-income ratio / avg. interest rate / avg. level of liquid asset.

Calibrating the Wealth Distribution

- Use the portfolio adjustment cost function (Kaplan and Violante ECTA 2014, Kaplan, Moll, Violante).
- Recall in KMV: $\chi(d, a) = \chi_0|d| + \chi_1|d/a|^{\chi_2}$
- Choose $(\rho, \kappa, \chi_1, \chi_2, \chi_3)$ to match fraction of individuals at the borrowing constraint, with negative wealth and mean liquid/illiquid assets.
- **Bayer, Born and Luetticke**: use wedge of interest rate between deposits and debt, and probability of portfolio rebalance to match ratio of liquid-illiquid, share of borrowers.

- **Even better** $\Rightarrow$ we can also target the aggregate MPC.
- Auclert, Rognlie, Straub (2023): target the MPC over the wealth distribution.
- **Problem:** In Brazil there are little data on wealth, MPC is even worse.
- What data there is in the BCB to calibrate these models?
 - ▶ Share borrowers?
 - ▶ Avg. value of liquid assets?
 - ▶ Fluctuation in credit card?

Cross-Sectional Identification in Macro

Building Empirical Evidence

- Having theoretically sounding models in macro are nice but we still need empirical evidence to back-up our results.
- Macroeconomists have traditionally used time-series to identify fiscal and monetary shocks (among other). Often relies on some type of identified VAR:
 - ▶ Structural restrictions, i.e., restrictions of response of variables to shocks;
 - ▶ Sign-restrictions;
 - ▶ Narrative approach, e.g., wars, oil shocks, identified tax changes;
- **Last 20 years:** tons of development in the applied micro literature $\Rightarrow$ macro researchers should embrace the causal-revolution!

Table 10: Econometric Methods and Data Types Over Time

Year Year	Methods		Data				
	Time series	Applied micro	Micro data	Time series	Cross section	Panel	Proprietary
1980	75	25	22	89	8	3	13
1990	62	38	28	70	14	16	32
2000	58	42	28	54	8	38	30
2006–10	46	54	41	42	13	45	41
2016–18	35	65	56	34	10	56	52

Note: The figures are the shares, expressed as percentages, of econometrics-based articles in the *JME* and *JMCB*, plus the E-designated articles in the five general-interest journals. The method and data attributes are defined in section 3.2. The 2006–10 figures use data from 2006, 2008 and 2010; and the 2016–18 figures use data from 2016, 2017 and 2018.

Cross-Sectional (Regional) Identification

- A particular useful approach is to explore the cross-sectional/regional exposure to an identified aggregate shock.
- **Example:**
 - ▶ Industry-specific international shock (aggregate time-series shock) affects disproportionately places with high-shares of the industry;
 - ▶ Monetary policy might affect low-wealth region differently than high-wealth.
- **Issues:**
 - ▶ Still need identification (narrative/instrument/diff-diff): is the monetary policy/international shock exogenous?
 - ▶ The region-individual must be exposed **before** the shock happens.
 - ▶ How to go from cross-sectional estimates to aggregate?

Example: Mian & Sufi (2014, ECTA)

- Impact of “housing net worth” on non-tradable employment at the county-level.
 - ▶ weaker household balance sheet $\Rightarrow$ demand shocks $\Rightarrow$ decline in real activity;

$$\Delta \log E_i^{NT} = \alpha + \eta \Delta \text{HNW}_i + \varepsilon_i$$

where $\Delta \text{HNW}_i = (\Delta p_{06-09}^{H,i} \times H_{2006}^i) / \text{NW}_{2006}^i$.

- **Aggregate Shock:** 2007-2009 housing prices' collapse;
- **Cross-sectional Heterogeneity:** county i housing as a share of net-worth.

Example: Autor, Dorn and Hansen (2013, AER)

- Effect of rising Chinese import competition on US employment.

$$\Delta L_i^M = \alpha + \beta \Delta IPW_i + \gamma X_i + \varepsilon_i$$

where i denote region and j denote industry:

$$\Delta IPW_i = \sum_j \frac{L_{ij,1990}}{L_{j,1990}} \frac{\Delta M_j}{L_{i,1990}}.$$

- ▶ **Aggregate Shock**: 1990-2007 change in imports from China by industry (ΔM_j);
- ▶ **Cross-sectional Heterogeneity**: initial differences in regional industry specialization.
- ▶ **Instrumental variables**: Chinese imports by other high-income countries.

Cross-Sectional Identification

- This type of regional identification is known as **Bartik** instrument or **shift-share**. Good references on the econometrics behind:
 - ▶ Goldsmith-Pinkham, Sorkin, and Swift (2020, AER), Borusyak, Hull, and Jaravel (2022, ReStud);
- **Main Problem:** in macro sometimes we want to know the aggregate impact of treatment, but the regression only tells you what happens in some regions relative to others;
 - ▶ The true aggregate relationship is different than the cross-sectional because of GE effects, spillovers, etc.
 - ▶ This is known as **the missing intercept problem**.
 - ▶ Example: it could be that trading with China increased aggregate employment, but it did *relatively less* in exposed regions $\Rightarrow$ the regression will say $\uparrow \Delta IPW \rightarrow \downarrow L^M$.

The Aggregation Issue

- **Problem:** (micro-) spillovers?
- **Example:**
 - ▶ Bad shock (treatment) in a region induces employment migration to a region that did not receive the shock;
 - ▶ Regression identifies differences in employment response from shock between regions $\Rightarrow$ migration increases the difference $\Rightarrow$ estimated β is larger;
 - ▶ But aggregate employment did not change as much, it just reallocated from one region to the other.
- **Stable Unit Treatment Value (SUTVA) Assumption:** treatment of one unit does not affect outcomes of non-treated units.
- If this assumption does not hold our regression estimates will be biased.

The Aggregation Issue

- **Problem:** General equilibrium effects (macro spillovers)?
- **Example:**
 - ▶ Good shock (treatment) in a region induces more consumption of tradable goods;
 - ▶ Price of the tradable good increases;
 - ▶ Control regions reduce consumption;
 - ▶ **Regression:** good shock increases consumption! but aggregate consumption did not really increase...
- Good theory tells you what the price transmission should look like.

The Aggregation Issue

- **Problem:** Endogenous responses?
- **Example:**
 - ▶ Bad output shock (treatment) in a region should not (alone) induce response by the monetary policy;
 - ▶ If we are interested in the total impact of the shock in the aggregate, we may want to consider the reaction of the monetary authority...
 - ▶ Feature or bug? Maybe we want to know effect of policy or shock separate from effect of policy response.
- Be aware of locally identified tax changes, federal government rebalancing budget, and any interactions between local and federal government.

The Aggregation Issue

- Does that mean we should not use this evidence? NO!
 - ▶ The micro evidence can help us to build better models...
 - ▶ if the empirical evidence rejects your model go back to the drawing board.
 - ▶ Look for the empirical evidence regarding the key elasticities of your model!
- The solutions to the aggregation issues are really problem dependent.
 - ▶ Estimate spillovers;
 - ▶ Use theory to guide general equilibrium/partial equilibrium effects;
 - ▶ Match causal moment in structural model (this is a type of indirect inference);